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WHY DO "100-YEARS CRISIS" HAPPENS EVERY 10 YEARS?

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MODELING NATURE

FULLER – 1914

- **RETURN PERIOD**

This deterministic equation is commonly considered to be the source of the interpretation of the return period as:

- time T until the occurrence of a T-year event,
- which can mean, for example, 100 years until the occurrence of a 100-year flood.

$$Q_T = CA^{0.8}[1 + 0.8\log_{10}(T)](1 + 2A^{-0.3})$$

- **50 YEARS OF OBSERVATION**

Fuller conducted hydrological observations in the second half of the 19th century. Fuller conducted statistical analyses of flood frequency in the USA.

- **USEFUL STATISTICAL TOOL**

This concept is very popular mainly because it is a simple statistical tool derived from engineering practice. For example, engineers working on flood control are interested in the expected time interval in which an event of a given magnitude will occur, which gives the basic definition of the return period.

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WHAT IS A CATASTROPHE?

QUESTIONS

Is a "out of nowhere" river flood also a flood?

Can we look at floods using statistics?

Is the same definition of catastrophe used in economics?

FLOOD

Flood is the temporary covering by water of land not normally covered by water, caused by an excessive amount of water from rivers, the sea or intense rainfall.

Source: EU Floods Directive (2007/60/EC): "temporary covering by water of land not normally covered by water"

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HYDROLOGY – STATISTICAL VIEW

- **HAZEN - LOGARITHMIC SCALE**

Hazen proposed a graphical approach to the analysis of flood data. He proposed plotting maximum flow values on a logarithmic scale, which de facto meant using the lognormal distribution.

- **PROBABILITY DISTRIBUTION?**

Hazen himself, in 1921, questioned the choice of the lognormal distribution in favor of a three-parameter distribution, accounting for the empirically observed possibility of skewness in logarithmized data. Then in 1924, Foster proposed the use of the Pearson type III distribution as the best representative of the phenomenon of maximum flows. However, it was later corrected to the form of log-Pearson type III, maintaining the assumption of a logarithmic distribution of flow maxima.

- **EXTREME VALUE THEORY (1928)**

At roughly the same time, Extreme Value Theory was being developed [Fisher, R. A., & Tippett, L. H. C., 1928], which introduced three types of extreme value distributions: type I – Gumbel, type II – Fréchet, type III – Weibull.

- **LOG-PEARSON TYPE III**

Ultimately, however, since 1967, when the approach was standardized, the log-Pearson type III distribution has been the official standard for flood frequency modeling in the United States. In general, the most commonly used theoretical distributions in flood frequency modeling in the United States are:

- Generalized Extreme Value (GEV),
- log-Pearson type III (LP3), and
- the three-parameter lognormal distribution (LN3).

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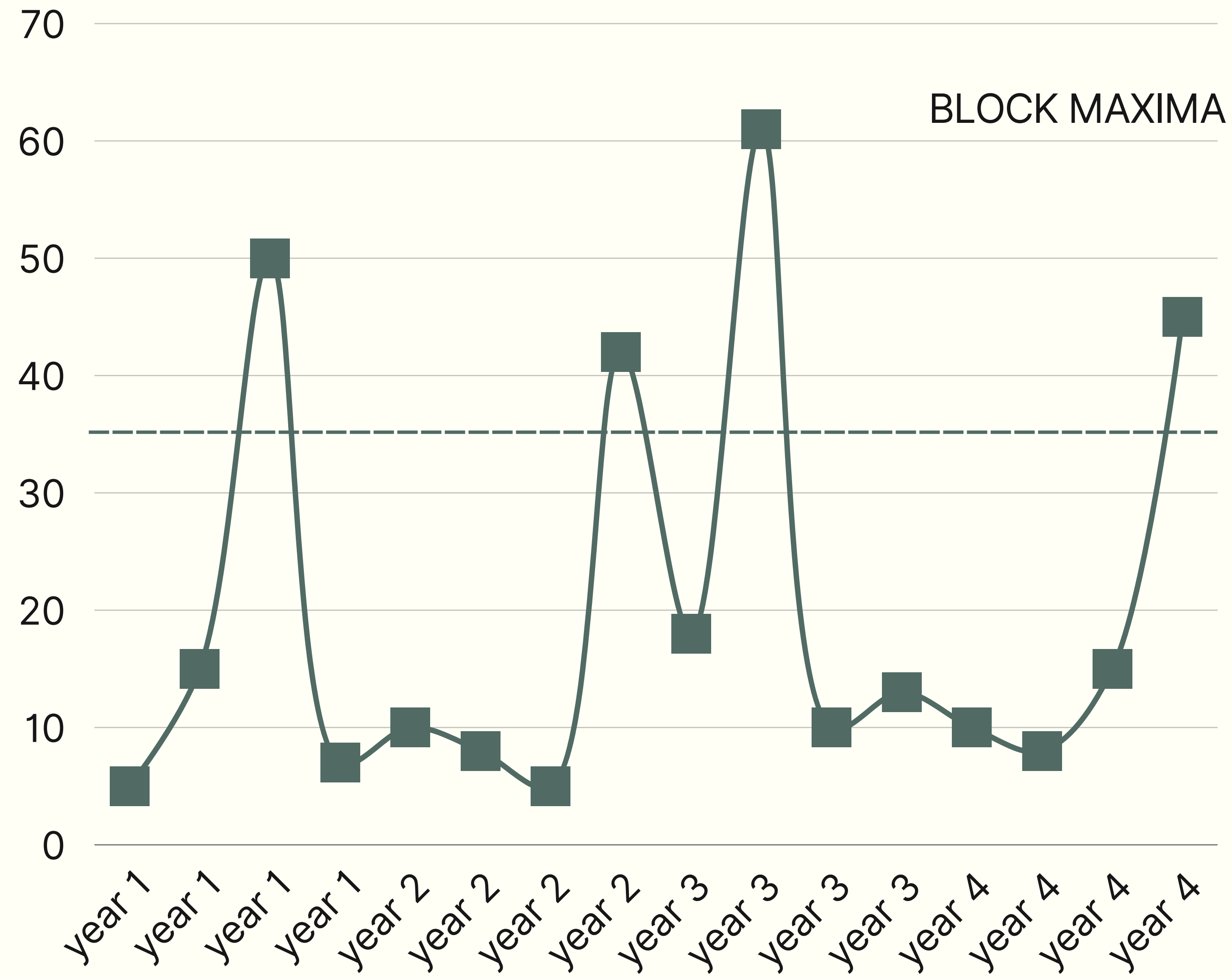
MODEL ANNUAL MAXIMUM



• NATURAL CYCLE

The foundation for such reasoning lies in the assumption of the repeatability of processes in certain closed cycles. In the case of events occurring in nature, such a natural cycle is the year. It should therefore be expected that during the year there is a moment of intensification of the phenomenon, and it is then that we will record its maximum for that period.

• BLOCK MAXIMA → 1/100 YEARS

The adoption of annual maxima as the modeled observation suggests an intuitive interpretation of the 0.99 quantile of such a distribution of maxima, as a frequency of once every 100 years of annual maxima observations.

EXCEEDANCE STATISTICS

- **PROBABILITY OF EXCEEDANCE**

$$P(X > x_p) = 1 - p$$

which should be interpreted as a 1-p chance of exceeding the value. For the quantile p = 0.99 we speak of an event with a 0.01 probability of exceeding the value located at that point of the distribution.

- **RETURN PERIOD**

$$T = \frac{1}{1 - p}$$

This means that for annual maximum flow data, an event that has a chance of exceeding a certain value in any given year with probability p = 0.01 (i.e., with a chance of 1 in 100 observations) can be called a 100-year flood (or generally a T-year flood). It is worth noting that the return period T does not imply that an event with such a T-year index (e.g., a 100-year event) will occur exactly every T years, but rather that each year there is a probability of 1/T of exceeding the T-year flood level.

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NO OF EXCEEDANCE

No of exceedance	Period T=100 100-years event Probability	Period T=1000 1000-years event Probability
0	36%	36%
1	36%	36%
2	18%	18%
3	6%	6%
4	1%	1%

- **EXPECTED NO OF EXCEEDANCE**

Binomial or Poisson distribution.
This means that the **expected number of exceedances in period T is 1**,
but
the **actual number of exceedances comes from a probability distribution and can take values: 0, 1, 2, 3**, etc.

- **TABLE - INTERPRETATION**

It is worth noting that there are exactly the same chances (36.8%) of one 100-year event occurring as of no 100-year event occurring in a period of 100 years. It can also be seen that there is a 6% chance of 3 hundred-year events occurring within 100 years. Similarly, there is a 36.8% chance of not observing any 1000-year event in 1000 years, the same chance of one such event in that period, and also a 6% chance of 3 thousand-year events in a period of 1000 years.

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RETURN PERIOD

k (years)	Period T=100 100-years event P(K=k)	Period T=100 100-years event P(K<k)
1	1%	1%
10	0.91%	9%
50	0.61%	39%
69	0.50%	50%
100	0.37%	63%
300	0.05%	95%

- **TIME TO 1ST EXCEEDANCE**

Instead of determining the expected number of exceedances in a period of T years, one can derive a formula for the time to the first exceedance, that is, the probability that the first exceedance at a given level will occur in year k. This can be written as follows, as a geometric distribution.

- **TABLE - INTERPRETATION**

A 100-year event will occur on average once in 100 years, and the mean waiting time for such an event is 100 years. However, it may happen that such a 100-year event does not occur at all within 100 years, or that several such events occur within that period. The first occurrence of a 100-year event can happen as early as after one year (1.00% chance), within the first 10 years (9.56% chance), within 50 years (39.5% chance), and within the studied 100-year period the first 100-year event has only a 63% chance of appearing, although within 300 years it should appear with a probability of 95%.

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RELIABILITY

• RELIABILITY FOR 100-YEARS EVENT

For T=100 and p=0.01, the reliability of the system over n=10 years is 90.4%, while the risk of failure is 9.6%. Within the next 10 years (in year 1, 2, 3, ..., or 10) we have a 9.6% chance that an extreme event with a return period of T=100 years (a 100-year event) will occur, a 90.4% chance that such an event will not occur in that period, and therefore that the system will maintain its resilience within that time horizon.

For a period of n=50 years we have a 39.5% chance of the event occurring and a 60.5% chance of it not occurring, and for n=100 years we have a 63.4% chance of the 100-year event occurring and a 36.6% chance of it not occurring, thus representing the reliability of the system.

• CHANCES OF SURVIVAL

A system designed for a "100-year flood":

- has only a 60.5% chance of surviving the next 50 years,
- which means there is almost a 40% chance that within those 50 years an event will occur that will destroy the system,
- or if we extend the required survival time to 100 years, we have only a 36.6% chance of survival.

• INCREASING LEVEL OF PROTECTION

Redesigning the system to be resilient to, for example, a 1000-year event, and therefore a more extreme one. With such a system design:

- we have a chance of surviving a period of 50 years with a probability of 95.1%,
- 100 years with a probability of 90.4%.

Such a resilience system, ensuring longer survival (or at least a greater probability of longer survival) will of course be significantly more costly.

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RELIABILITY FOR REAL PROBLEMS

- **DIFFERENT PARAMETRIZATION**

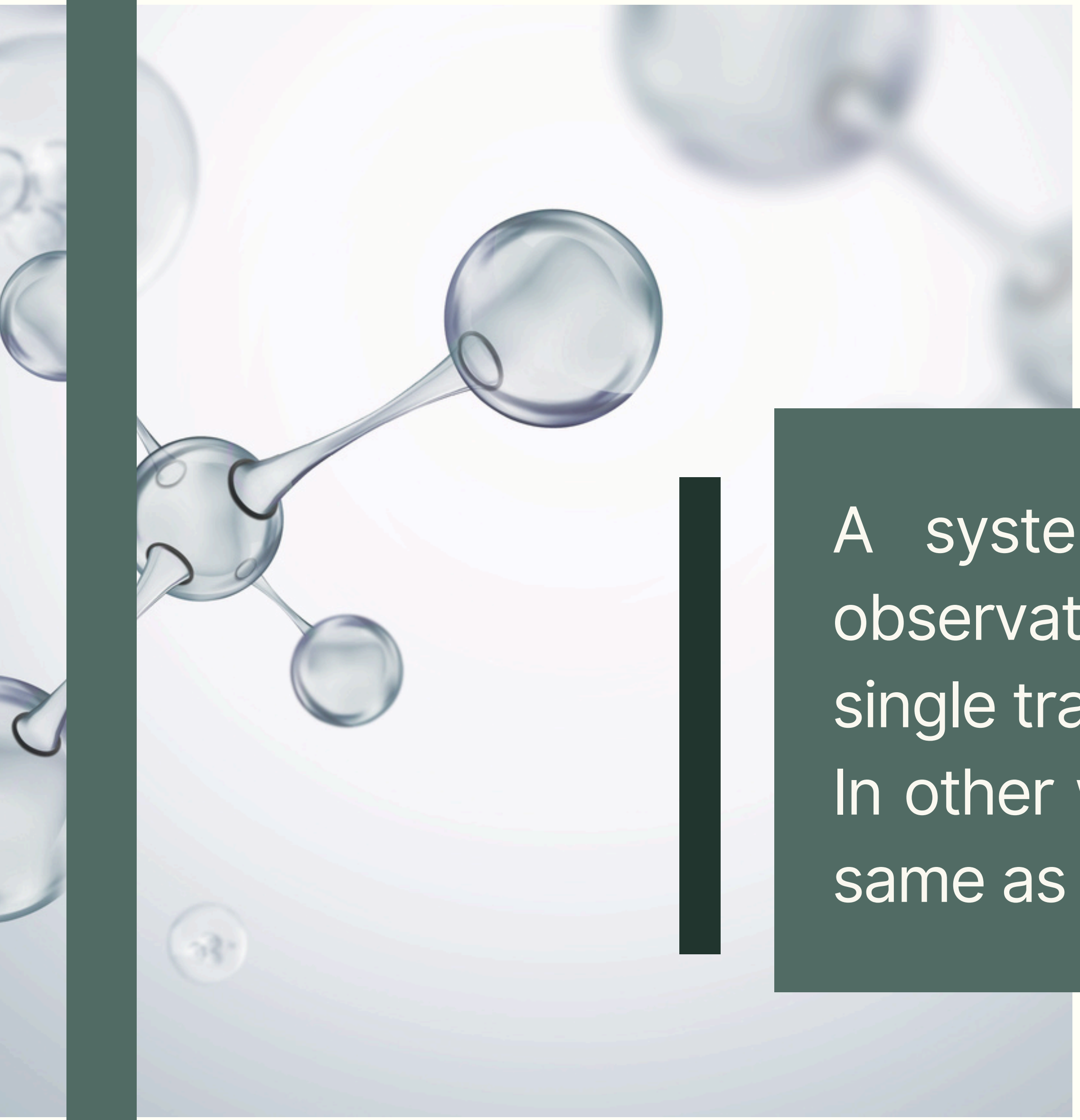
The results indicate differences in the designed safety levels measured by the reliability index.

- A structure in a seismically active area will have a 98% chance of surviving 50 years, which can be read as only 2 out of 100 buildings not surviving that period. This is because the level of p has been set at 2475 years, meaning an extremely unlikely, 2475-year event.

- Meanwhile, only 61 out of 100 flood barriers built at the same level of resilience will survive a period of 50 years. Here the safety level is defined as a 100-year event.

Domain	Reliability for planing period n years	Expected return period (T-years)
Earthquakes	98% n=50	2475
Nuclear power plants	95% n=21	409
Flood insurance	75% n=30	105
Flood infrastructure	61% n=50	100

Source: Based on Read and Vogel (2015)



ERGODICITY

A system is ergodic when the average across many parallel observations at one point in time gives the same result as observing a single trajectory over a long period of time.

In other words — one person living a hundred years experiences the same as a hundred people each living one year.

STATIONARITY - I.I.D.

- **INDEPENDENT**

Successive observations are independent of each other

- **IDENTICALLY DISTRIBUTED**

All observations have the same distribution:

- the same mean and
- the same variance

- **I.I.D. IN HYDROLOGY**

In hydrology, this should be understood as meaning that in the observed period (e.g. T=100 years) we assume constant conditions, and therefore the invariability of the distribution and its variance (unchanged probability p), as well as the independence of successive observations coming from such a distribution of flows.

- **GUMBEL'S ARGUMENTATION**

*"In order to apply any theory, **we have to suppose that the data are homogeneous**, i.e., that **no systematical change of climate and no important change in the basin have occurred within the observation period** and that no such changes will take place in the period for which extrapolations are made. **It is only under these obvious conditions that forecasts can be made."***

- **I.I.D - COMMON ASSUMPTION**

The hydrological models that have been in use for the last 100 years in the United States, as well as models commonly used in other countries, were based on the assumptions of: (i) stationarity of the process, and also (ii) independence of observations.

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STATIONARITY IS DEAD

Stationarity Is Dead: Whither Water Management?

P. C. D. Milly,[1*] Julio Betancourt,[2] Malin Falkenmark,[3] Robert M. Hirsch,[4] Zbigniew W. Kundzewicz,[5] Dennis P. Lettenmaier,[6] Ronald J. Stouffer[7]

MILLY ET AL., 2008

“The envelope of natural variability has been exceeded”
“Stationarity is dead.”

WHAT DOES MILLY SAY?

“Systems for management of water throughout the developed world have been designed and operated ***under the assumption of stationarity.****”*

„we assert that stationarity is dead and should no longer serve as a central, default assumption in water-resource Risk assessment and planning.”

"Projected changes in runoff during the multidecade lifetime of major water infrastructure projects begun now are large enough to push hydroclimate beyond the range of historical behaviors.".

„Stationarity cannot be revived”

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CRITIQUE OF THE PROBABILISTIC APPROACH

- ## MORE STRUCTURAL VIEW

Merz R., Blöschl G., 2008:
“If more light is to be shed on the probabilities of hydrological events, then **it will have to come from more information on the physics of the phenomena, not from more mathematics.**”
“the hydrological literature has focused too much on solving technical estimation problems (distribution fitting, parameterization methods), neglecting the wealth of hydrological knowledge.”

- ## FUNDAMENTAL QUESTION

does the concept of 1/1000 still make sense?

- ## STATISTICS IS LIMITED

Dawdy Griffis
point to the insurmountable limitations of traditional statistical models. Despite the many improvements that have been made to such models over time, they still remain a **black box, closed to knowledge of the real physical processes taking place.**

- ## PHILOSOPHICAL SHIFT

[**Milly et al., 2008**
aptly, it seems, point to the necessary change in modeling philosophy. **The model should not be the central point of reference**, but merely an aid in synthesizing observations:

"Modeling should be used to synthesize observations; it can never replace them."

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100-YEARS FLOOD MISINTERPRETATION

- **PIELKE (1999)**

"The 'base flood' is more commonly known as 'the 100-year flood' and is ***probably the most misunderstood floodplain management term"***

"Misconceptions about the meaning of the term creates obstacles to proper understanding of the flood problem and, consequently, the development of effective responses."

First, there is general confusion among users of the term about what it means. ***Some use the term to refer to a flood that occurs every 100 years****, as did the Midwestern mayor who stated that 'after the 1965 flood,* ***they told us this wouldn't happen again for another 100 year****s' (IFMRC, 1994, p. 59).*

*Public confusion is widespread: A farmer suffering through Midwest flooding for the second time in three years complained that '****Two years ago was supposed to be a 100-year flood, and they're saying this is a 75-year flood, What kind of sense does that make****? You'd think they'd get it right'*

Correct definition should be: *„...flood that has a one percent chance of being exceeded in any given year."*

A 100-year flood is not a natural concept arising from the built-in characteristics of nature. It is rather a **definition constructed by scientists and engineers**, developed on the basis of historical observations, which should be dynamic due to changing natural conditions as well as emerging new observations. **Meanwhile, regulatory definitions are not so easy to change.**

Pielke pointed to the need to **redefine the way flood risk is described**. He advocated for measuring the phenomenon through **the concept of reliability, referring to a calculation indicating a 26% chance of a 100-year flood within 30 years**, which should be the most useful measure for determining the level of protection.

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100-YEARS FLOOD MISINTERPRETATION

- **READ, VOGEL (2015)**

*“**The consequences of misinterpreting** the expected waiting time to an extreme event **do not only impact the physical system but also impact perceptions of individuals** interacting with the floodplain.”*

*“**Floodplain residents generally expect to be protected for a particular average return period and have little understanding of the true risk of flooding during their lifetimes.**”*

*“**Risk communication is itself a complex issue**, as empirical research suggests that obstacles to understanding risks from natural hazards often require strategies deeper than presenting facts, such as using tactics to dispel heuristics and preconceived mistaken theories.”*

They conclude that the return period and communicating risk through it (e.g. a 100-year flood occurring once in 10 years) is problematic even under conditions of stationarity. It does not communicate the true risk of catastrophe within the planning horizon.

Furthermore, stakeholders, including politicians, engineers, insurers, and citizens, **misinterpret the return period precisely as "once in 100 years"**.

Under conditions of non-stationarity, things are even worse. The return period no longer depends on a constant probability p — now p changes over time, and this means instability in the interpretation of the so-called 100-year flood. An event defined in this way may in fact be a 200-year flood (p=0.0005) in the year 2020, an 83-year flood (p=0.012) in the year 2050, and a 56-year flood (p=0.018) in the year 2070.

At the same time, **they state that the concept of reliability, despite the difficulties of deriving it under non-stationary conditions, is still, as in stationary conditions, a better practical tool**. This is because it accounts for the variability of conditions (non-stationarity) and provides answers regarding safety within the assumed planning horizon.

REGULATOR: 1/1000 YEARS

- **BIS BCBS 1998: MARKET RISK 1/100 YEARS**

"In calculating the value-at-risk, a 99th percentile, one-tailed confidence interval is to be used."

- **BIS BCBS 2005: CREDIT RISK 1/1000 YEARS**

*"The confidence level is fixed at 99.9%, i.e. an institution is expected to suffer losses that exceed its level of tier 1 and tier 2 capital **on average once in a thousand years**. This confidence level might seem rather high. However …"*

- **JORION 2007 - PUBLICATION**

*"The Basel Committee requires banks to hold sufficient tier 1 and tier 2 capital to cover unexpected losses over a 1-year horizon at the 99.9 percent confidence level. This seems very conservative. It means that a bank would fail, **on average, once in a thousand years.**"*

- **MILONAS, VON VUUREN 2024 - PUBLICATION**

*"… a confidence level of 99.9% is sufficient since losses that exceed the ULs are predicted **to occur only every 1000 years** because the PDs used are annual PDs."*

- **GENEST, BRIE 2013 - PUBLICATION**

*"That's why the confidence level is set at 99.9%. It means a bank can have losses that exceed its tier 1 and tier 2 capital **on average once in a thousand years.**"*

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1/1000 YEARS

1/1000 YEARS

Answer at the level of the 99.9% quantile understood as once in 1000 years: **it is possible** (over such a long period of time a complete reshuffling of the structure of economic forces in the world is possible)

1/1000 ONE-YEAR SCENARIOS

Answer at the level of the 99.9% quantile understood as once in 1000 scenarios: **it is NOT possible** (it is obvious that even under the most favorable circumstances it is not possible to dominate the global car market within a single year)

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1/1000 YEARS IS SO SIMPLE?

Regulator (BIS, BCBS) did NOT define fundamental concepts:

- lack of stationarity assumptions
- lack of interpretation: what does 1/1000 years mean
- expected return period not defined
- reliability not defined

HOW CAN WE INTERPRET THIS?

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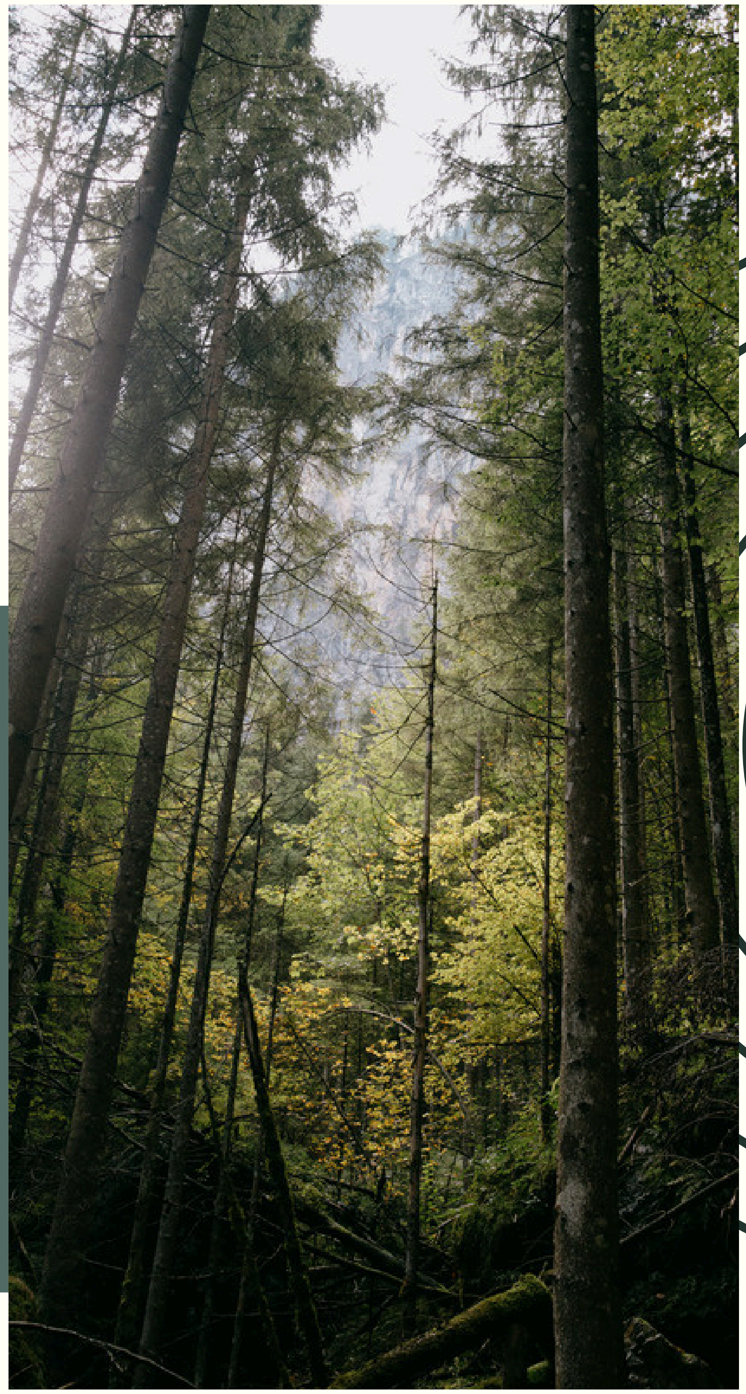
MODELING NATURE IS SIMPLE

SOCIAL PHENOMENA ARE NOT STABLE

Social phenomena are inherently dynamic and subject to constant change, as they are shaped by human behavior, cultural shifts, technological progress, and political transformations. Unlike the laws of physics, social processes do not follow fixed rules — what was true for one generation, one era, or one economic system may be completely irrelevant in a different historical or cultural context.

THE LAWS OF PHYSICS ARE STABLE

The laws of physics have remained unchanged throughout the entire history of the universe — gravity, thermodynamics and electromagnetism operate the same way today as they did billions of years ago. Unlike statistical models based on historical data, physical laws do not require the assumption of stationarity, as they describe the fundamental mechanisms governing natural processes regardless of changing conditions.



THE WORLD AS RANDOM

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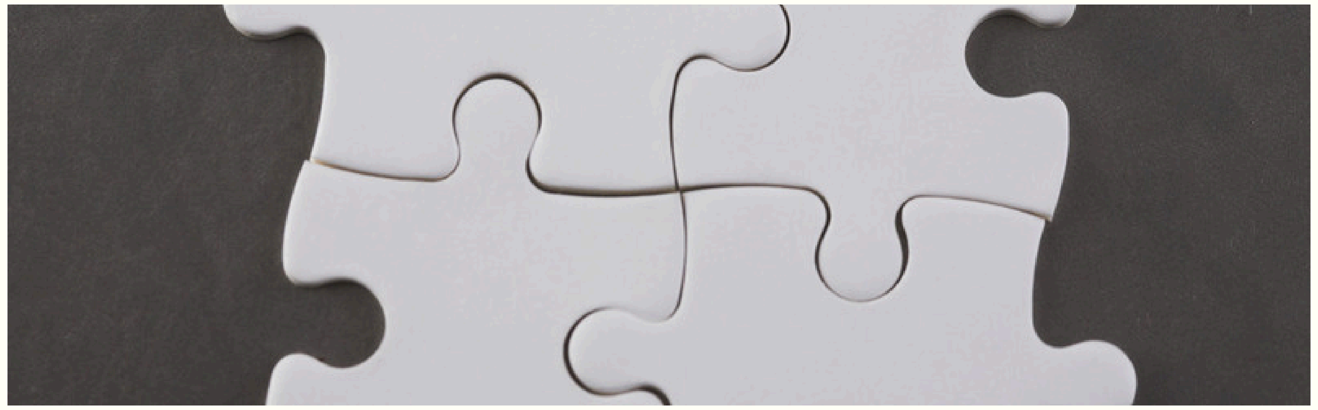
WHAT IS THE WORLD LIKE?

DETERMINISTIC

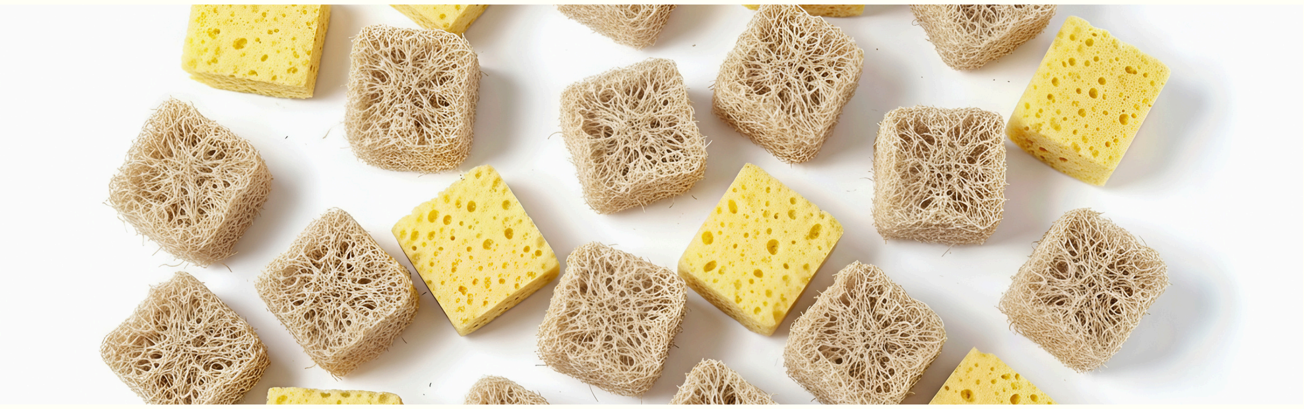
Leibniz - 17th century
Everything that exists has a sufficient reason for its existence — the principle of sufficient reason. Nothing happens without a cause. Nothing is random. Chance is merely an expression of our ignorance.



STOCHASTIC → RANDOM

Maxwell - 19th century
randomness is the proper description of reality

Heisenberg - 19th century
The world is random



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PROBABILITY CALCULUS

MILESTONES

- Pascal and Fermat in the 17th century — probability calculus as the language of describing uncertainty.
- De Moivre 1733, Laplace 1812 — the normal distribution as a universal description of the sum of random events.
- Gauss 19th century — the method of least squares, measurement error as normal. The world is normal.

WHAT IS IT?

Probability calculus was originally developed in the 17th century to solve problems related to games of chance, when Pascal and Fermat began analyzing the mathematics of dice and card games. Over time, it evolved into a universal tool for describing uncertainty and randomness in the natural world, economics, and science.

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DETERMINISTIC WORLD

MILESTONES

- **Leibniz — rationalist determinism.** Leibniz in the 17th century said that everything that exists has a sufficient reason for its existence — the principle of sufficient reason. Nothing happens without a cause. Nothing is random. Chance is merely an expression of our ignorance — if we knew enough, we would see the necessity of every event.
- **Newton — mechanical determinism.** Newton gave this image a concrete machine. The laws of motion, gravity — knowing the position and velocity of every body at a given moment, you can calculate its position at every future moment. The universe as a clock. Perfect, predictable, necessary.
- **Laplace — absolute determinism**, the most radical Laplace at the beginning of the 19th century formulated this most sharply of all. Imagine a mind that knows the position and momentum of every particle in the universe at a given moment. Such a mind could calculate the entire future state of the universe and the entire past.

DETERMINISTS

They believed that the universe operates like a perfect clockwork mechanism, in which every event is the inevitable consequence of preceding causes governed by fixed laws of nature.

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A KEY TURN MAXWELL AND BOLTZMANN

WHAT DID THEY SAY?

- Maxwell and Boltzmann did not say — trajectories do not exist. They did not say — the world is not deterministic.
- They said something more subtle — trajectories may exist, but it is the wrong question. The wrong level of description.

The shift was not in ontology — in how the world is. It was in epistemology — in what questions are worth asking.

THE AUTONOMY OF THE MACROSCOPIC

- **Before them science had one ideal of explanation** — find the causes, derive the trajectory, predict. One level of description considered the only true one.
- **After them science had two levels of description that are equally true and equally scientific**.
- Microscopic — trajectories, causes, mechanics.
- **Macroscopic — distributions, probabilities, statistics.** !
- And crucially — the macroscopic level is not inferior, it is not an approximation, it is not a temporary substitute for when we do not know the trajectories. **It is a complete and autonomous description of reality.**

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GALTON & PEARSON

GALTON AND PEARSON 1880-1900

Statistics as the language of social sciences
This was a crucial bridge — between Maxwell's statistical physics and the sciences of man.
Galton and Pearson said: man is also statistical matter.

THEIR CONTRIBUTION

Galton — Darwin's cousin — was **the first to systematically apply the normal distribution to human phenomena**.
Height, intelligence, heredity. He showed that human traits have distributions, that they can be measured, compared, and correlated.

Karl Pearson developed this into an entire statistical school. The correlation coefficient, regression, statistical tests. He created a language with which social phenomena could be described just as a physicist describes gases.

Pearson created a system — a family of distributions that encompasses many different shapes under one mathematical umbrella.

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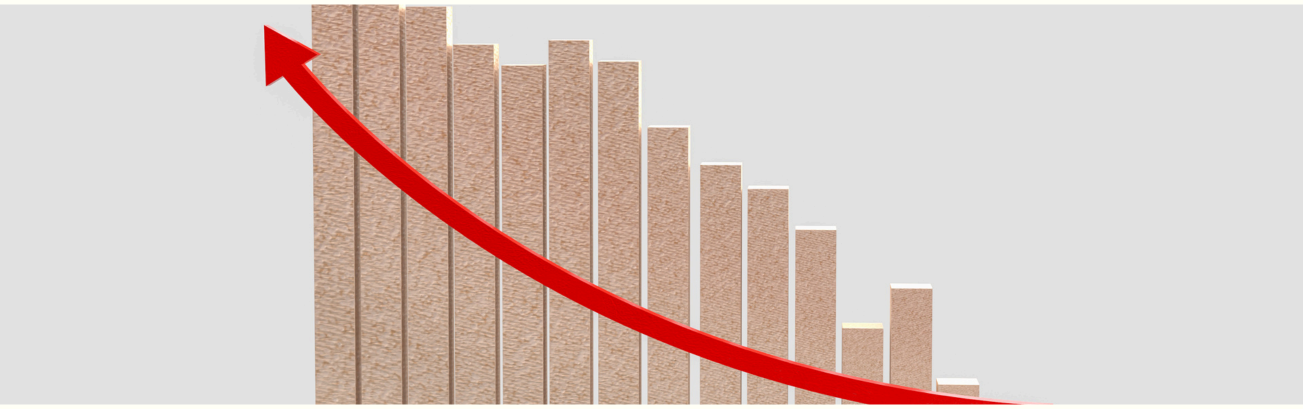
WHAT IS A PROBABILITY DISTRIBUTION?

STRUCTURAL APPROACH

The earlier ones — Gauss, Poisson, Laplace — discovered specific distributions **starting from specific physical or mathematical problems.**

INSTRUMENTALIST APPROACH

Pearson reversed the direction.
Instead of — I have a problem, what distribution describes the mechanism — he said — **I have data, fit the distribution that describes it best.**
He waged 30-year battles with Ronald Fisher.



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PEARSON – INSTRUMENTALIST

Pearson was a positivist in the philosophical sense — and this was no coincidence. He wrote "Grammar of Science" where he stated outright: **science does not discover reality, science builds economical descriptions of phenomena**. **Formulas are mental shortcuts, not descriptions of deep structure**. For Pearson a distribution was a curve that you fit to data. The question "why does the data have this shape" was for him the wrong question — metaphysical, not scientific.

SCIENTIFIC INSTRUMENTALISM

WHAT IS SCIENTIFIC INSTRUMENTALISM

Scientific instrumentalism is a philosophical position that says — **scientific theories are not claims about reality. They are tools for predicting and organizing experience.**

A theory does not describe how the world is. A theory works or it does not work — it is useful or it is not useful.

CONTRAST WITH REALISM

A realist says — electrons really exist. The theory of electrons is true because it describes real objects.
An instrumentalist says — I do not know whether electrons really exist and that question is not scientific. The theory of electrons is good because it allows me to predict experimental results. The electron is a useful mental construct.

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BROADER CONTEXT

IS THE WORLD SUBJECTIVE?

- ### CRACKS IN RATIONALISM

By the end of the 19th century, five great cracks had accumulated in the rationalist project.

Schopenhauer and Nietzsche — the foundation of reality is not reason but will, drive, perspective.
Darwin — man is an accident, history has no teleological meaning.
The crisis of mathematics — even formal certainty is a construct, not a discovery.
Freud — the knowing subject is irrational, governed by the unconscious.
The religious crisis — the traditional foundation of meaning and values collapsed.

- ### THE TURNING POINT 1890-1910

All these cracks matured simultaneously and exploded into a new cultural movement.

If there is no objective truth — **paint the impression, not the object**. **Impressionism.**
If consciousness is a stream — **write stream of consciousness, not a linear narrative**. **Woolf, Proust, Joyce.**
If the foundation is drive and emotion, not reason — **music must express emotion, not structure**. **Strauss, Debussy, Stravinsky**.
If geometry is a construct — **reality can be described from multiple perspectives simultaneously**. **Cubism, Picasso**.
If values are created not discovered — **the artist creates their own world of meaning. Expressionism.**

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THE WORLD OF ECONOMICS

IS THIS WORLD OBJECTIVE OR SUBJECTIVE?

- ### 19TH CENTURY - RATIONALISM

The economists of the 19th century were children of the Enlightenment.
Ricardo, Mill, Marx, Marshall — **all believed that the economy has a structure that can be discovered through reason**.
Value has objective causes. Economic history has a direction.
Man is homo economicus — rational, predictable, utility-maximizing.
This was a project parallel to Newton's physics. Just as Newton discovered the laws of motion — economists wanted to discover the laws of value, the laws of the market, the laws of history.
Marshall at the turn of the 19th and 20th centuries — the mathematatization of economics, supply and demand curves, market equilibrium as a clockwork mechanism. Elegant, rational, predictable.

- ### PRACTICAL CONSERVATISM

While the entire intellectual culture around it was going through a great crisis of certainty — mainstream economics held on to the rationalist project.
Partly because it had a practical task — to advise governments, banks, and firms. **Unquantifiable uncertainty is intellectually honest but practically useless.**

This fit the era. The 19th century was industrialization, railways, telegraph, medicine, empire. Progress was visible, measurable, real. **It was easy to believe that history has a direction and that reason governs it**.
The great crisis of this narrative came from outside — not from philosophy, but from history. World War I 1914-1918 was the end of rationalist optimism in culture — **the rationalist narrative of progress is empty.**

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BACHELIER - 1900

"THÉORIE DE LA SPÉCULATION" - RANDOM WALK

• WHAT CAME BEFORE BACHELIER?

Before Bachelier, stock market prices were understood through two traditions.

- The first tradition — fundamental. A stock price reflects the value of a company. The value of a company depends on profits, assets, and prospects. A good analyst who understands the company can predict whether the price will rise or fall.
- The second tradition — technical. Prices have patterns — trends, resistances, supports. A good observer who reads charts can predict price movements.

Both traditions assumed that prices have a predictable structure that can be discovered through analysis. And both traditions assumed that **randomness — if it exists at all — is a disturbance of the signal**. Noise that obscures the true movement of the price.

• HE REVERSED THIS PERSPECTIVE

Bachelier did not only say — prices are random.

He said — **randomness is a property of informational equilibrium**.

This is something deeper. **Prices are random not because the market is chaotic — but because it is efficient.**

A paradox — the more rational the market, the more random the prices.

And he said — **the mathematics appropriate for describing prices is not classical mechanics with trajectories and forces. It is probability calculus and stochastic processes**.

This is the same breakthrough as Maxwell and Boltzmann in physics. **Instead of trajectories — distributions. Instead of determinism — probabilism**.

Bachelier transferred the probabilistic revolution from physics to economics. Thirty years after Maxwell. In the same intellectual climate.

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NORMAL DISTRIBUTION

The central limit theorem states that the sum of many independent random shocks converges to a normal distribution — which made it the natural model for asset returns, treated as the accumulated effect of countless small pieces of information arriving in the market

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UTILITY THEORY

BERNOULLI 1738 — MARGINAL UTILITY

Ytility grows more slowly than money. Logarithmically. Each additional zloty gives you less satisfaction than the previous one.
Man is naturally risk-averse. He prefers a certain 50 zloty over a fifty-fifty bet between zero and one hundred zloty — even if the expected value of the bet is exactly 50. Because the logarithm of certainty is higher than the expected value of the logarithm of the bet.

VON NEUMANN AND MORGENSTERN (1944)

Publication: "Theory of Games and Economic Behavior"

If a person's preferences satisfy a few simple conditions of rationality, then there exists a utility function that they maximize.
Von Neumann and Morgenstern gave economics an elegant model of a rational agent — **a maximizer of expected utility**. This was homo economicus in its probabilistic version.
The market aggregates the decisions of rational utility maximizers and produces an efficient price.

Why then not model human behavior as rational agents?

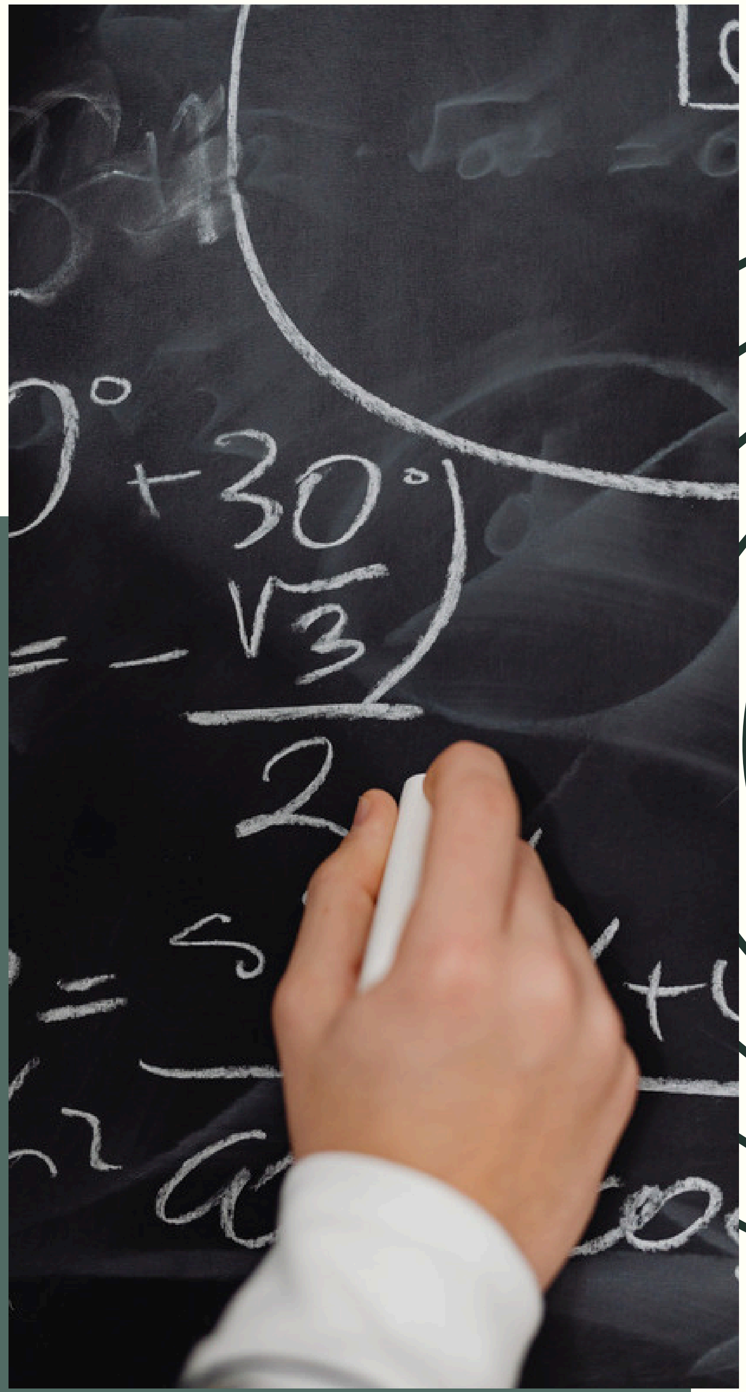
MASSACHUSETTS INSTITUTE OF TECHNOLOGY

MATHEMATIZATION OF ECONOMICS

Massachusetts Institute of Technology — was not a traditional place for economics. It was a place for engineers and physicists.
When economics began to mathematize in the 1940s and 1950s — MIT became the natural center of this revolution. Because it had a mathematical and engineering tradition that was lacking at traditional economics departments such as Harvard or Chicago.

PAUL SAMUELSON - FOUNDING FATHER

"Foundations of Economic Analysis" — 1947. This is perhaps the most important book in the history of 20th century academic economics.
Samuelson showed that all of economics can be written as optimization problems with equilibrium conditions. The consumer maximizes utility. The firm maximizes profit. The market reaches equilibrium where supply equals demand.
All of this was known before. But Samuelson showed that all these problems have the same mathematical structure. And that they can be analyzed with one set of tools — differential calculus, linear algebra, optimization theory. **The formalist revolution.**



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MASSACHUSETTS INSTITUTE OF TECHNOLOGY

MIT was a factory that produced two things simultaneously.
Formalist tools — models, equations, optimization — which dominated 20th century economics
and
the doctoral students who spread these ideas around the world: Solow, Modigliani, Bernanke, Blanchard, Draghi, Acemoglu, Krugman

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PRAGMATISM

MILTON FRIEDMAN (1953)

“The Methodology of Positive Economics"

"the only relevant test of the validity of a hypothesis is comparison of its predictions with experience"

"Truly important and significant hypotheses will be found to have assumptions that are wildly inaccurate descriptive representations of reality"

Scientific theories are simplifications. Always. The map is not the territory. The model is not reality. If the model were as complex as reality — it would be useless.

The proper question is — does the theory predict accurately? Are its implications consistent with observations?

INTELLECTUAL SHIELD

He gave economists philosophical permission to use models with unrealistic assumptions without intellectual guilt. Before Friedman, critics could attack economic models by saying — your assumptions are false. Homo economicus does not exist. Markets are not perfectly competitive. Information is not perfect. After Friedman, an economist could respond — I am not interested in whether the assumptions are realistic. I am interested in whether the model predicts. And that is all. This was an intellectual shield that protected models from criticism of their foundations.

NEXT STAGES

OF BUILDING THE FINANCIAL SYSTEM

- ### PORTFOLIO THEORY (1952)

Harry Markowitz — Portfolio Theory (1952)

Markowitz gave the financial system its most fundamental tool: Modern Portfolio Theory (MPT). His key insight was that investors should not evaluate assets in isolation, but in terms of how they interact within a portfolio. By combining assets with imperfectly correlated returns, an investor can reduce risk without sacrificing expected return — leading to the concept of the efficient frontier.

This translated the intuition of diversification into a precise mathematical framework using variances and covariances, becoming the foundation for asset pricing, risk management, and portfolio optimization used by every major financial institution in the world today.

- ### EFFICIENT MARKET HYPOTHESIS (1970)

Eugene Fama — Efficient Market Hypothesis (1970)

Fama gave the financial system its most influential theoretical framework: the Efficient Market Hypothesis (EMH). His key insight was that in a well-functioning market, prices fully reflect all available information at any given moment. This means that no investor can consistently outperform the market through analysis or timing — because any known information is already priced in.
This translated Bachelier's earlier intuition about random prices into a rigorous academic framework, distinguishing three forms of market efficiency — weak, semi-strong, and strong. It became the intellectual foundation for index investing, passive portfolio management, and the entire debate about active versus passive strategies that shapes asset management to this day.
And this built the ideological foundation for everything that followed. Markets are efficient — prices are correct — probabilistic models are appropriate.

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NEXT STAGES

OF BUILDING THE FINANCIAL SYSTEM

- ### STRUCTURAL CREDIT RISK MODEL

Robert C. Merton (1974)

Merton said — an option is a redundant asset. Its payoff can be exactly replicated through dynamic management of a portfolio composed of stocks and bonds.

If you can replicate an option — then its price must equal the cost of replication. Otherwise there is arbitrage — you can earn without risk.

Bankruptcy as an option, credit risk is quantifiable.

- ### BLACK AND SCHOLES — SYNTHESIS

Fischer Black and Myron Scholes (1973)
"The Pricing of Options and Corporate Liabilities", Journal of Political Economy, 1973.

Black and Scholes derived a partial differential equation for the price of an option. The assumptions were as follows. The stock price follows geometric Brownian motion — lognormal distribution of returns, constant volatility. The market is continuous — one can trade at any moment.

Option pricing through no-arbitrage, volatility as the only risk parameter. Volatility is constant and known.
Traders began using the formula with textbooks and calculators. Texas Instruments released a calculator with the Black-Scholes formula built in.
The model created a convention. And the convention created market reality.

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KEY MOMENT

BLACK-SCHOLES

A single number was seen by bank boards, by regulators, by rating agencies as an objective, scientific, precise measure of risk.
Knightian uncertainty — unquantifiable, irreducible, fundamental — was converted into quantifiable risk through a single chain of assumptions.

SYNTHESIS

- Black-Scholes did not say — every risk is priceable.
- But it built a machinery in which that claim was an inevitable consequence. Geometric Brownian motion for prices — gave normal distributions of returns. Normal distributions of returns — gave measurable volatility as the complete description of risk.
- Measurable volatility — gave a replicating portfolio for every instrument.
- A replicating portfolio — gave an objective price for every instrument.
- An objective price — gave an objective measure of risk.
- An objective measure of risk — gave a single number for the entire portfolio.

Now we have: closed-form formulas, a single number for risk, "we are in control".

NEXT STAGES

OF BUILDING THE FINANCIAL SYSTEM

- **VASICEK MODEL (1987)**

The Vasicek model (1987) decomposes credit risk into systematic risk — common to all borrowers — and idiosyncratic risk — specific to each borrower, which diversifies away in a large portfolio. Asset returns are assumed to follow a normal distribution, which became the basis of Basel II capital requirements. A financial crisis is an extreme quantile of this distribution — and the normal distribution severely underestimates the probability of such tail events.

- **J.P.MORGAN - VAR (1994)**

Value at Risk (VaR) was developed and popularized by JP Morgan in the early 1990s and released publicly as the RiskMetrics framework in 1994. The model collapses the entire risk of a portfolio into a single number — the maximum loss over a given time horizon at a given confidence level — assuming a normal distribution of returns. Its simplicity made it universally adopted, and Basel II embedded it as the official regulatory measure of market risk. The critical flaw is identical to Vasicek — the normal distribution severely underestimates tail events. VaR tells you where the threshold is, but says nothing about the size of losses beyond it. During the 2008 crisis, losses across major institutions exceeded their VaR estimates by multiples — events the model classified as occurring once in millions of years happened on consecutive days.

NEXT STAGES

OF BUILDING THE FINANCIAL SYSTEM

• J.P.MORGAN - CREDIT METRICS (1997)

CreditMetrics was groundbreaking for several reasons.

- First — it standardized the language. Before CreditMetrics every bank measured credit risk differently. After CreditMetrics there was a common framework — rating migrations, transition matrix, Credit VaR.
- Second — it legitimized the quantitative approach to credit risk. Credit risk stopped being an art of credit assessment — it became a science measurable through models.
- Third — it opened the door to securitization. If you can measure the risk of a loan portfolio — you can price it. If you can price it — you can sell it. CreditMetrics was the intellectual foundation for CDOs.

Alternative models: KMV, CreditRisk+, CreditPortfolioView

• COPULA - DAVID LI

David Li (2000)
"On Default Correlation: A Copula Function Approach"

Li read CreditMetrics and saw a structural problem.

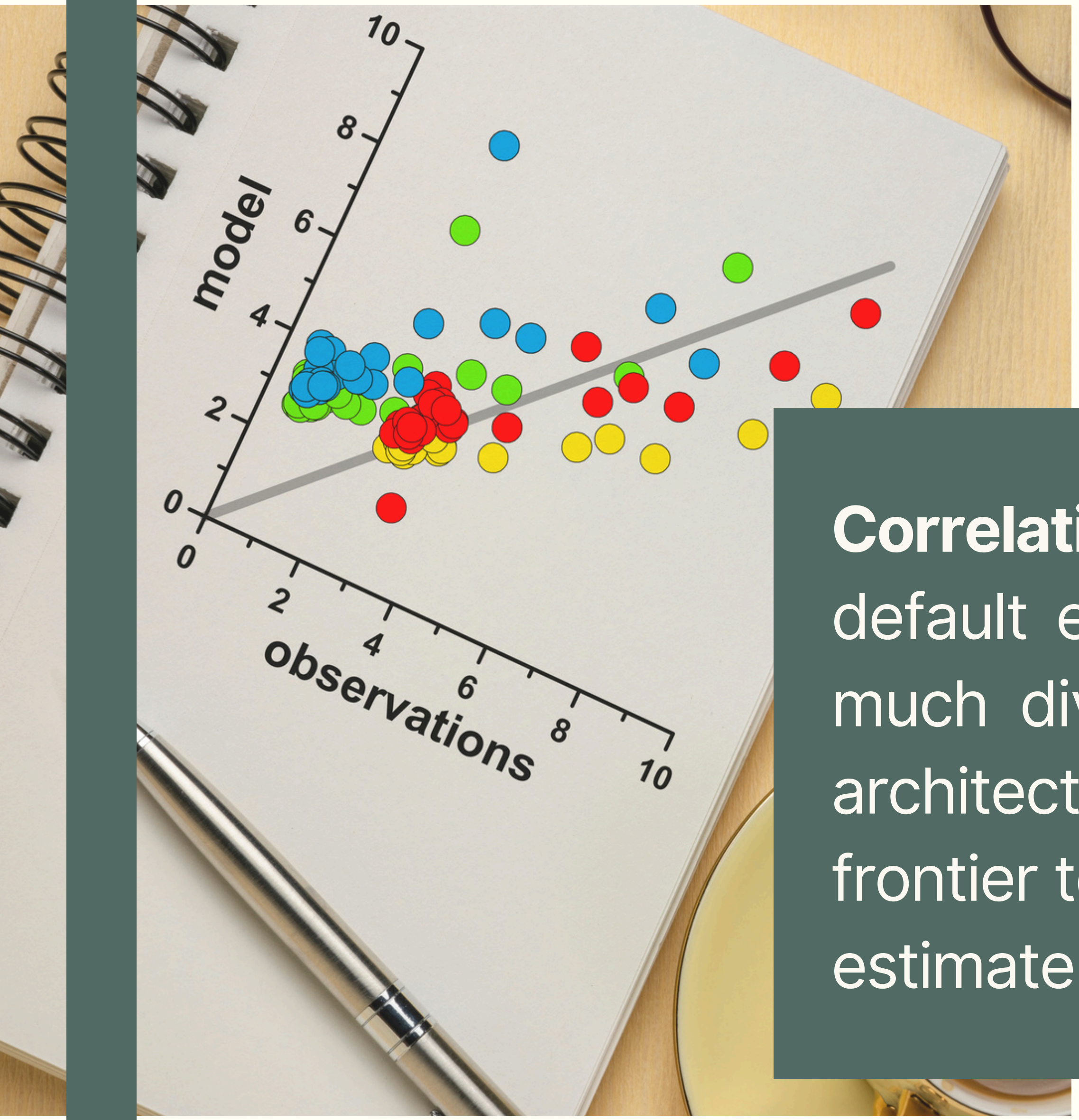
CreditMetrics modeled correlations through asset value correlations — through the Merton model.

This required equity data and assumptions about the capital structure of the firm.

Li said — there is a more elegant solution. Instead of modeling the mechanism of correlation — I model directly the dependence between times to default.

This is the key shift. From mechanism to statistics. From the question of why to the question of how much together.

DEPENDENCE



Correlation in finance measures the degree to which asset prices or default events move together — a parameter that determines how much diversification actually reduces risk in a portfolio. The entire architecture of modern quantitative finance, from Markowitz's efficient frontier to Vasicek's credit model to Li's copula, is built on the ability to estimate and model these correlations from historical data.

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NOBEL PRIZE

- **1970 - PAUL SAMUELSON**

Mathematization of economics, "Foundations of Economic Analysis"

- **1985 - FRANCO MODIGLIANI**

Life-cycle theory and corporate finance

- **1990 - HARRY MARKOWITZ**

Portfolio theory — efficient frontier

- **1997 - ROBERT MERTON, MYRON SCHOLES**

Option pricing, credit risk model, Black-Scholes formula

- **2013 - EUGENE FAMA**

Efficient Market Hypothesi

THEY BUILT COHERENT SYSTEM

Together they built a coherent system. The financial world is describable through stochastic mathematics:

- Risk is quantifiable.
- All financial instruments can be priced through no-arbitrage and the lognormal distribution.

This is intellectually beautiful. And philosophically it is a project of converting Knightian uncertainty into Knightian risk — through mathematics.

FUNDAMENTAL ASSUMPTIONS

- **THE WORLD AS RANDOM**

No matter what phenomena are, we can see them as random

- **ASSET PRIZES - BROWNIAN MOTION**

Geometric Brownian motion became the default language for describing asset prices — elegant, tractable.

- **FORMALIZATION OF ECONOMICS**

The formalization of economics transformed it from a descriptive social science into a mathematical discipline built on optimization, equilibrium, and probability. What began with Samuelson's equations and Markowitz's matrices ended with models that could price any instrument, quantify any risk, and reduce the complexity of financial reality to a set of tractable parameters.

- **FRIEDMAN'S PRAGMATISM**

Milton Friedman argued that the realism of a model's assumptions is irrelevant — what matters is whether the model predicts well. This pragmatist position gave quantitative finance its philosophical license: it did not matter whether markets truly followed geometric Brownian motion, as long as the Black-Scholes formula produced useful prices.

- **UNCERTAINTY IS MEASURABLE**

Uncertainty is measurable — this is the fundamental assumption underlying the entire edifice of modern quantitative finance. Every model, from Black-Scholes to VaR to the Gaussian copula, rests on the premise that the unknown can be expressed as a probability distribution with estimable parameters.

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FUNDAMENTAL ASSUMPTIONS

- **NORMAL DISTRIBUTION**

The normal distribution describes price movements — small changes are frequent, large changes are rare, and extreme events are virtually impossible.

- **NORMAL AGGREGATE**

And the aggregated distribution is also normal — by the central limit theorem, the sum of many small random shocks converges to a normal distribution regardless of their individual shapes.

- **CRISIS - QUANTILE ON NORMAL**

A financial crisis is an extreme quantile of the normal distribution — an event so far in the tail that the model assigns it a probability of once in millions of years, yet it happens in reality far more often.

- **RISK DIVERISIFICATION**

Markowitz showed that by combining assets with imperfectly correlated returns, an investor can reduce portfolio risk without sacrificing expected return — the mathematical proof that diversification works.

- **VASICEK CORRELATION**

Vasicek assumed that correlations between borrowers are driven by a single common factor — the state of the economy — and that in a large portfolio, idiosyncratic risks cancel out, leaving only this systematic component.

- **LI CORRELATION**

Li modeled the dependence between times to default directly using a Gaussian copula — bypassing the need to model the underlying mechanism, and reducing correlation to a single parameter.

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REGULATOR

ENTER THE REGULATOR

The regulator stepped in in 1988 for three reasons.

- Herstatt 1974 showed that the bankruptcy of one bank immediately hits others — the need for international coordination.
- The debt crisis of 1982 showed that banks are undercapitalized relative to the risks they take.
- Japanese banks showed that unequal capital requirements create unequal competition. Basel I attempted to fix these three problems through a common definition of capital and minimum requirements.

THE BASEL COMMITTEE - CREDIT RISK (BASEL I - 1988)

The Basel Committee — chaired by Peter Cooke of the Bank of England, hence sometimes called the Cooke ratio — published its agreement in July 1988 with a simple and limited ambition.

- First — define capital: standardize what counts as regulatory capital into Tier 1 and Tier 2.
- Second — introduce risk weights: assign assets to risk categories so that capital requirements reflect asset riskiness.
- Third — set a minimum capital level: 8% of risk-weighted assets, of which at least 4% must be Tier 1.

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BASEL COMMITTEE

CAPITAL REQUIREMENTS FOR MARKET RISK (1996)

• PROBLEM TO SOLVE

Basel I of 1988 covered only credit risk — the risk that a borrower will not repay.

But banks also had increasingly large trading books. Stocks, bonds, currencies, derivatives that they bought and sold — not to hold to maturity, but to trade and profit from price differences.

These trading books were exposed to market risk — that asset prices would move adversely before the bank had time to sell them.

Basel I had no capital requirements for market risk. This was a regulatory gap. And this gap was growing — because banks were rapidly expanding their trading books throughout the 1980s and early 1990s.

• INTERNAL MODELS OF MARKET RISK

The 1996 Amendment to Basel I introduced capital requirements for market risk — and with it came a critical decision. Regulators had two options:

- a standardized approach with fixed risk weights defined by regulators, simple but crude; or
- internal models, allowing banks to use their own Value at Risk models to calculate capital requirements.

Regulators chose internal models, accepting the banks' argument that their own models were more precise, capturing correlations, diversification, and actual volatility of specific instruments. Why?

- banking lobby pressure
- intellectual attractiveness of quantitative models
- faith in technological progress: regulators believed that banks which had invested in advanced risk systems genuinely understood risk better than any simple regulatory formula could capture.

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BASEL COMMITTEE

NEW CAPITAL ACCORD - BASEL II (2004)

- **PROBLEM TO SOLVE**

Basel I said — every corporate loan has a 100% risk weight. Hold 8% capital.

But 8% on a loan to General Electric and 8% on a loan to a small pizzeria on the edge of bankruptcy — that is absurd. These two loans carry completely different risk.

Basel II said — the capital requirement should reflect the actual risk of a specific loan. Not the broad category it belongs to.

- **INTERNAL MODELS OF CREDIT RISK**

Basel II gave banks a formula that converted these four parameters into a capital requirement.

The formula IRB was based on the Vasicek model. One common systematic factor Z plus the idiosyncratic risk of each loan. The formula looked as follows in simplified form.

Risk weight =
function(PD, LGD, M, correlation with systematic factor)
Capital requirement = 8% × EAD × Risk weight

The higher the PD — the higher the risk weight — the more capital required.
The higher the LGD — the higher the risk weight.
The longer the maturity — the higher the weight.

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ALL OF THIS WORLD CAN BE SEEN IN BANKING MODELS

- **ECL MODELS - EXPECTED LOSS**

Expected value models measure the level of provisions — that is, those credit losses that occur in an average year.

- **CAPITAL ADEQUACY MODELS (0.999)**

Capital adequacy models are aimed at measuring the level of protective capital — the level of own funds that are designed to absorb crisis losses.

- **NORMAL REALITY**

The concepts are based on the assumption of normality of phenomena, meaning the absence of dependence, identical distribution, and the absence of a dominant factor.

- **REGULATORY APPROVAL**

The regulator in Basel II enabled risk modeling through internal models. **Additionally it popularized these methods — the entire market calculated this way.**

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IN 2004 WE WERE SAFE

1/1000 YEARS SAFETY (0.999)

The regulator defined safety at the level of 1/1000 years. To construct such a concept it used the mechanics of the normal distribution. It did not take into account the problems with interpreting such a measure.

This was not the only problem.

THREE AREAS OF SAFETY - BASEL II (2004)

Credit risk — Basel II replaced crude risk weights with internal models based on PD, LGD, EAD, and maturity, making capital requirements reflect the actual risk of each borrower. In 2004 we believed that credit risk was finally measurable and properly capitalized.

Market risk — the 1996 Amendment allowed banks to use internal VaR models, embedding the belief that a single number at 99% confidence was a sufficient description of market risk. In 2004 we believed that volatility was stable and the normal distribution captured everything that mattered.

Operational risk — Basel II introduced capital requirements for operational risk for the first time, covering losses from failed processes, systems, and human error. In 2004 we believed that even the most elusive category of risk had been quantified.

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WE ARE SAFE BECAUSE ...

We see the world through mathematics and we understand it.
We can calculate risks and their probabilities.
We see it from above — risks are external, they are part of the world we observe, and we stand above it.
We know the **scale and probability of catastrophes: once in 1000 years.**